Ian Schillebeeckx

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Data science leader turned AI-native IC. With agentic tools, the line between managing humans and AI agents has blurred—delegating experimental design, evaluating results, and iterating on analyses feels the same whether through Slack or a terminal. Using Codex agents, I personally matched the output of the other five members of my team. Effective managers are the new super ICs.

Education

Washington University in St. Louis — St. Louis, MO
PhD, Computer Science (Machine Learning) – Fastest graduation in department history — *2012 – 2016*

Saint Louis University — St. Louis, MO
Honors BS, Computer Science — *2007 – 2011*
Honors BS, Applied Mathematics

Experience

Director of Enterprise Data Science and AI — January 2025 – Present
CareDx, Inc. (Medical Diagnostics, ∼750 employees) — *San Francisco, CA*

- Built company's first data science function outside R&D: embedded with Sales, Operations, and Finance as trusted analytical partner to CEO, CFO, and COO; proactively surfaced insights and opportunities driving customer engagement, retention, and revenue growth
- Ran incrementality and causal inference studies to measure efficacy of business interventions: applied causal DAGs, causal random forests, linear regression, difference-in-differences, and propensity matching to quantify marginal impact of sales processes on revenue; translated complex findings into actionable recommendations driving 10% revenue increase (∼$25M)
- Built forecasting system covering ∼75% of revenue: benchmarked ARIMA, Prophet, and custom objective functions, reducing forecast error from 6% to 0.9%; designed Databricks ETL pipelines for model training and inference; deployed self-service dashboards with automated monitoring enabling stakeholders to self-serve routine data and insights
- Built predictive churn model covering $270M revenue: novel piecewise linear changepoint detection + dynamic programming approach replacing 6-month-lag reactive detection with forward-looking predictions; Databricks ETL pipeline with automated anomaly detection triggering alerts on data refresh; customer segmentation insights drove proactive engagement strategies across territories
- Built tools for AI-driven productivity: designed agentic system generating personalized next-best-action recommendations per customer territory; built custom CLI tools (database access, SQL generation, schema introspection) enabling self-serve data workflows; improved practitioner productivity 50%
- Shipped 3 LLM applications: enterprise RAG app ($1M investment) for self-service knowledge retrieval; ProductLLM scoring Target Product Profiles against rubric on market potential and technical feasibility; PaperLLM generating original research manuscripts (12 produced, 2 submitted to journals)
- Drove company-wide Copilot adoption to 92% (∼750 employees), saving 8% of total hours annually; authored enterprise AI governance policy
- AI-native practitioner: built custom Codex tool stack with specialized database access, SQL generation, and schema introspection tools for data science team; drove adoption across broader 20-person Data and AI group (Statistics, Data Engineering, Bioinformatics, Data Science) through executive sponsorship and regular peer training sessions

Director of Data Science — January 2024 – January 2025
CareDx, Inc. — *San Francisco, CA*

- Led multidisciplinary data organization (14 people, $6M budget) spanning Bioinformatics, Statistics, and Data Science; established key metrics and measurement frameworks; fostered data-driven culture of experimentation and continuous learning
- Directed production ML models (LSTM, SVM, LASSO) with hands-on technical depth: end-to-end from development through deployment, monitoring, and governance; set standards for statistical rigor across the organization
- Oversaw enterprise analytics portfolio of ∼20 Tableau dashboards; built self-service analytics enabling cross-functional teams to derive insights independently; drove data quality, governance, and metric consistency

Head of Product and Data Science — December 2017 – January 2024
Cofactor Genomics (Oncology Diagnostics) — *San Francisco, CA*

- Owned end-to-end delivery of 2 production ML models (classification, regression): feature engineering from high-dimensional data, model optimization via bootstrap cross-validation, scalable cloud deployment with automated regression testing, CI/CD, and model monitoring

- Architected data platform from scratch: ML model repository (SageMaker), versioned training data (100s GBs), cloud pipelines (AWS Batch, +200% throughput, -25% cost), CI/CD (GitHub Actions)—self-served all underlying data and derived insights from them
- Designed and executed experimentation program across 12+ sites (~500 subjects): prespecified hypotheses, statistical power analysis, rigorous measurement of intervention efficacy informing business strategy
- Led multidisciplinary team of 10; defined product roadmaps and OKRs; partnered across Product, Engineering, and Commercial to translate complex problems into actionable analytical frameworks
- First author on Nature ML paper (2nd most downloaded in 2022); authored 3 AI patents (2 granted)

CEO & Founder May 2017 – August 2020
Where4Care (Technology Startup) *St. Louis, MO*

- Built product from zero: designed matching algorithm, full tech stack (Python/Django), cloud infrastructure; validated through 200-user study; hands-on end-to-end from data architecture through production deployment

Selected Publications & Patents

3 Artificial Intelligence Patents (2 granted) – Methods for prediction, classification, and data processing using ML/AI

ML-based Prediction Model – *Nature Scientific Reports, 2022* (2nd most downloaded paper that year)

12 Peer-Reviewed Publications – Applied ML, deep learning, experimentation *(CVPR, ECCV, Nature, ICCP, 3DV)*

Technical Skills

Causal Inference & Experimentation: A/B Testing, Causal DAGs (Back-Door Criterion, Do-Calculus), Causal Forests, Difference-in-Differences, Propensity Matching, Synthetic Controls, Instrumental Variables, Regression Discontinuity, Incrementality Analysis, Statistical Power Analysis

ML & Analytics: Classification (SVM, XGBoost), Regression (LASSO), Deep Learning (LSTM, PyTorch), Time Series Forecasting, Customer Lifecycle Analytics, Churn/Retention Modeling, Segmentation, Anomaly Detection, Model Monitoring

AI & Tools: LLMs, RAG (Embeddings, Vector Search), Agentic AI (Custom Tool Building), Self-Service Analytics, Data Storytelling (Decks, Memos, Dashboards)

Languages & Platform: Python (expert), SQL (expert), R, AWS (SageMaker, Batch), Databricks, Tableau, Streamlit, ETL/Data Pipelines, CI/CD, Docker, Git

GTM & Business: Sales/GTM Analytics, Marketing Measurement, Customer Success, Revenue Optimization, Cross-Functional Partnership (Sales, Marketing, Finance, Product, Engineering, Support, Executive)